

OpenStax Calculus Volume 3

Section 7.1 Exercises: Second-Order Linear Equations

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Classify each of the following equations as linear or nonlinear. If the equation is linear, determine whether it is homogeneous or nonhomogeneous.

1. $x^3y'' + (x - 1)y' - 8y = 0$
2. $(1 + y^2)y'' + xy' - 3y = \cos x$
3. $xy'' + e^y y' = x$
4. $y'' + \frac{4}{x}y' - 8xy = 5x^2 + 1$
5. $y'' + (\sin x)y' - xy = 4y$
6. $y'' + (\frac{x+3}{y})y' = 0$

For each of the following problems, verify that the given function is a solution to the differential equation. Use a graphing utility to graph the particular solutions for several values of c_1 and c_2 . What do the solutions have in common?

7. [T] $y'' + 2y' - 3y = 0$; $y(x) = c_1e^x + c_2e^{-3x}$
8. [T] $x^2y'' - 2y - 3x^2 + 1 = 0$; $y(x) = c_1x^2 + c_2x^{-1} + x^2 \ln(x) + \frac{1}{2}$
9. [T] $y'' + 14y' + 49y = 0$; $y(x) = c_1e^{-7x} + c_2xe^{-7x}$
10. [T] $6y'' - 49y' + 8y = 0$; $y(x) = c_1e^{x/6} + c_2e^{8x}$

Find the general solution to the linear differential equation.

11. $y'' - 3y' - 10y = 0$
12. $y'' - 7y' + 12y = 0$
13. $y'' + 4y' + 4y = 0$
14. $4y'' - 12y' + 9y = 0$
15. $2y'' - 3y' - 5y = 0$
16. $3y'' - 14y' + 8y = 0$
17. $y'' + y' + y = 0$

18. $5y'' + 2y' + 4y = 0$
19. $y'' - 121y = 0$
20. $8y'' + 14y' - 15y = 0$
21. $y'' + 81y = 0$
22. $y'' - y' + 11y = 0$
23. $2y'' = 0$
24. $y'' - 6y' + 9y = 0$
25. $3y'' - 2y' - 7y = 0$
26. $4y'' - 10y' = 0$
27. $36\frac{d^2y}{dx^2} + 12\frac{dy}{dx} + y = 0$
28. $25\frac{d^2y}{dx^2} - 80\frac{dy}{dx} + 64y = 0$
29. $\frac{d^2y}{dx^2} - 9\frac{dy}{dx} = 0$
30. $4\frac{d^2y}{dx^2} + 8y = 0$

Solve the initial-value problem.

31. $y'' + 5y' + 6y = 0, y(0) = 0, y'(0) = -2$
32. $y'' + 2y' - 8y = 0, y(0) = 5, y'(0) = 4$
33. $y'' + 4y = 0, y(0) = 3, y'(0) = 10$
34. $y'' - 18y' + 81y = 0, y(0) = 1, y'(0) = 5$
35. $y'' - y' - 30y = 0, y(0) = 1, y'(0) = -16$
36. $4y'' + 4y' - 8y = 0, y(0) = 2, y'(0) = 1$
37. $25y'' + 10y' + y = 0, y(0) = 2, y'(0) = 1$
38. $y'' + y = 0, y(\pi) = 1, y'(\pi) = -5$

Solve the boundary-value problem, if possible.

39. $y'' + y' - 42y = 0, y(0) = 0, y(1) = 2$
40. $9y'' + y = 0, y(\frac{3\pi}{2}) = 6, y(0) = -8$
41. $y'' + 10y' + 34y = 0, y(0) = 6, y(\pi) = 2$
42. $y'' + 7y' - 60y = 0, y(0) = 4, y(2) = 0$
43. $y'' - 4y' + 4y = 0, y(0) = 2, y(1) = -1$
44. $y'' - 5y' = 0, y(0) = 3, y(-1) = 2$
45. $y'' + 9y = 0, y(0) = 4, y(\frac{\pi}{3}) = -4$
46. $4y'' + 25y = 0, y(0) = 2, y(2\pi) = -2$

47. Find a differential equation with a general solution that is $y = c_1 e^{x/5} + c_2 e^{-4x}$.
48. Find a differential equation with a general solution that is $y = c_1 e^x + c_2 e^{-4x/3}$.

For each of the following differential equations:

(a) Solve the initial value problem.

(b) [T] Use a graphing utility to graph the particular solution.

49. $y'' + 64y = 0$; $y(0) = 3$, $y'(0) = -24$

50. $y'' - 2y' + 10y = 0$, $y(0) = 1$, $y'(0) = 13$

51. $y'' + 5y' + 15y = 0$, $y(0) = -2$, $y'(0) = 7$

52. (Principle of superposition) Prove that if $y_1(x)$ and $y_2(x)$ are solutions to a linear homogeneous differential equation, $y'' + p(x)y' + q(x)y = 0$, then the function $y(x) = c_1 y_1(x) + c_2 y_2(x)$, where c_1 and c_2 are constants, is also a solution.

53. Prove that if a , b , and c are positive constants, then all solutions to the second-order linear differential equation $ay'' + by' + cy = 0$ approach zero as $x \rightarrow \infty$. (Hint: Consider three cases: two distinct roots, repeated real roots, and complex conjugate roots.)